

Intermediate Microeconomics - Mathematical Approach (ECO 310): Spring 2026

Professor: Andrea Wilson (awilson@princeton.edu)

Class: Tues & Thurs, 10:40-12 in Lewis Library 120

Office Hours: (Tentatively) Mon 4-5 in JRR 230. You can also visit (but with lower priority) during my other class's office hours Wed 4-5, and are welcome to email me with questions or to make an appointment for another time.

Preceptors: Yinshan Shang (yinshans@princeton.edu) and Paroma Chowdhury (pr6553@princeton.edu)

- **NOTE: precepts begin in the second week of classes**

Textbook: No required textbook. (I can recommend one if you feel that extra reading would be helpful). Lecture slides will be posted on Canvas, and should contain everything you need to know.

Evaluation: Best 4 of 5 problem sets (20%), Exams 80%:

- you may have one 24h extension; for issues beyond this, please use it as your dropped one
- Exam weights: whichever is better of 30% midterm + 50% final, or 20% midterm + 60% final
- Exam timing:
 - the midterm will take place Thurs, March 5 (likely during class, some chance I make it online)
 - the final will be **in person** (by department mandate), on Tues May 12 4-7pm

Tentative Problem Set Schedule: • PS1: posted Mon Feb 2, due Thurs Feb 12

- PS2: posted Mon Feb 16, due Thurs Feb 26
- PS3: posted Mon March 23, due Thurs April 2
- PS4: posted Mon April 6, due Mon April 13
- PS5: posted Tues April 14, due Thurs April 23

Disability Accommodations: Students must register with the Office of Disability Services (ods@princeton.edu) for disability verification and determination of eligibility for reasonable academic accommodations. Requests for academic accommodations for a course need to be made to ODS at the beginning of the semester, or as soon as possible for newly approved students, and again at least two weeks in advance of any needed accommodations so that arrangements can be made.

Precepts and Class Attendance: Precepts will mostly go through additional practice problems. You are expected and encouraged to attend. But there is no explicit attendance grade, either for lecture or for the precepts.

Course Outline

Part 1. **Consumer Behavior (5 classes: Jan 27,29, Feb 3,5,10)**

- intro + preferences, utility
- rational choice; revealed preference
- budget-constrained utility maximization; demand functions
- income and substitution effects; more properties of demand

Part 2. **Uncertainty and Financial Assets (3.5 classes: Feb 12,17,19,24)**

- expected utility
- applications
- paradoxes and extensions

Part 3. **Equilibrium (1.5 classes: Feb 24,26)**

- market demand; equilibrium
- taxes; externalities

Midterm Week: No Class and/or Review March 3, Midterm March 5

Spring Break: week of March 9

Part 3: **Firm Behavior; Markets (6 classes)**

- Firm Behavior and Supply (March 17,19)
 - competitive markets: SR and LR
 - monopoly
- General Equilibrium (March 24,26)
 - exchange economies
 - double auctions
 - rational expectations
- Matching (March 31, April 2)

Part 4: **Game Theory (4 classes)**

- NE in one-shot games (April 7,9)
- SPNE in dynamic games (April 14,16)

Part 5. **Information Economics (2 classes: April 21,23)**

- Bayesian games + intro to mechanism design
- adverse selection and moral hazard